

Md. Ahasan Ahamed

Micro/Nano Systems Engineer | Biosensing & Diagnostics | Microfluidics | Electrical, Mechanical & Biomedical Integration
Bloomington, IN · +1 945-214-5328 · mja6561@psu.edu · LinkedIn | Google Scholar | ResearchGate | ORCID
U.S. Permanent Resident, authorized to work in the U.S. without sponsorship

Highlights

- Ph.D. candidate in Electrical Engineering, Penn State (expected August 2026); research focus on single-molecule nanopore sensing and point-of-care device fabrication, with expertise in isothermal amplification (RPA, LAMP, CRISPR), reaction kinetics, biophysics, and ML-driven signal analysis.
- **20+** peer-reviewed publications (8 first/co-first author, 4 second author) in *ACS Sensors*, *Analytical Chemistry*, *Analytical Chimica Acta*, and *Biosensors & Bioelectronics*; **254 citations**, h-index **8** (Google Scholar).
- Disease areas: Alzheimer disease, HIV, Mpox, and SARS-CoV-2; developed multiplex detection assays and early-stage biomarker prediction workflows.
- Contributed to NSF ECCS Award #2603909 (\$494K, 2026–2029) and NIH proposal preparation; established and operationalized the Guan Lab at Indiana University, including instrument procurement, qualification, and in-house maintenance.
- **6+** years of teaching and mentoring experience (Teaching Assistant, Guest Lecturer); mentored undergraduate, M.S., and junior Ph.D. students.

Education

- **Ph.D. in Electrical Engineering** 2022 – 2026
 - The Pennsylvania State University, University Park, PA CGPA: 3.72
 - Dissertation: Integrating Recombinase Polymerase Amplification with Nanopore Sensing for Point-of-Care Detection of Infectious Diseases.
 - Major: Optics, Semiconductor device design, Bio-electronics System
 - Supervisor: Weihua Guan, PhD
- **M.S. in Mechanical and Production Engineering** 2020 – 2022
 - Konkuk University, Seoul, South Korea CGPA: 4.25 & 96.5%
 - Thesis: Pre-programmed Microdroplet Generator to Control Wide-ranging Chemical Concentrations.
 - Major: Advanced fluid dynamics, Micro/Nano-Fluidics, Advanced Numerical Analysis, Robot Kinematics
 - Supervisor: Sung-Jin Kim, PhD
- **B.Sc. in Mechanical Engineering** 2009 – 2014
 - Bangladesh University of Engineering and Technology (BUET), Bangladesh CGPA: 3.67 & 77.4%
 - Major: Thermal & Fluid Sciences, Renewable Energy, Instrumentation & Control System
 - Thesis: Electricity Generation from the Compression of Speed Breakers.
 - Supervisor: Arif Hasan Mamun, PhD

Skills

- **Technical Skills**
 - **Experimental Techniques**
 - Wet lab: Assay development (CRISPR, RPA, LAMP, PCR), microfluidic chip fabrication and bonding, solid-state nanopore fabrication, soft lithography, digital droplet quantification.
 - Fluorescent data analysis, gel electrophoresis, Nanodrop quantification, digital analysis.
 - **Instrumentation & Characterization**
 - Laser system: Universal Laser Systems (VLS3.60ES), Coherent OBIS (488/550nm 20/100mW), Sutter Instrument P-2000.

- Additive & Subtractive Manufacturing: MakerBot Method X, Asiga UV Max X43, Ultimaker 3.0, Pocket NC V2-10 Desktop 5-axis CNC, Formlabs Form 3.
- Microscopes: Nikon Eclipse Ti-U, AmScope Inverted, confocal systems, general optical microscopes (Nikon C-DSD230, Olympus CK40, SZ51).
- Vacuum & Plasma: HFS Vacuum Oven, Edwards Scroll Vacuum Pump, Welch Dryfast vacuum pump, Plasma Etch PE-25, FemtoScience Cute-1MP.
- Pressure Systems: PX-309 series, Eve Flow series, ELVESYS AF1 200, Fluigent Flow EZ flow controller, Fluigent FLPG+ low-pressure generator, Chemyx Fusion 100 syringe pump.
- Fluorescence Detection: Infinite 200 Pro Plate Reader, BIORAD GelDoc Go, T100 & CFX96 Touch.
- Signal & Power Systems: Trek 623B, Rigol DS4024, DG4162, DP832, Axopatch 200b, Element e1b, DAQ systems.
- Cameras: Novitec USB camera u-Nova20C, Sony CCD camera HK5CCD-s, Dino USB camera.
- **Embedded Systems & Device Integration**
 - Hardware-software integration and circuit build-up for portable POC devices; sensor-to-electronics integration, embedded readout, real-time data acquisition.
 - NVIDIA Jetson Orin edge-computing integration with Python-based APIs for PC-to-hardware device communication; PCB design, schematic and physical layout.
- **Computational & Data Analysis**
 - MATLAB/Simulink, Python, C++, ImageJ, Origin, Excel, PowerPoint, Word, COMSOL Multiphysics, PLECS, TCAD Sentaurus (MOSFET/BJT).
 - Machine Learning & Statistical Analysis: DSP, feature extraction, predictive modeling in biophysics, statistical process control.
- **Design & Engineering**
 - SolidWorks, AutoCAD, Inventor, LabVIEW (Automation & Control), Arduino microcontroller.
 - LaTeX (Academic Writing), Shotcut, InShot (Video Editing).
- **AI & Agentic Workflows**
 - Anthropic Claude and ChatGPT for engineering and analysis; agentic AI in VS Code (Claude Code, Codex) for code generation, automation, and agent-based task execution.
- **Soft Skills**
 - **Scientific Writing** – Peer-reviewed articles, NSF/NIH grant proposals.
 - **Project Management** – Led teams, managed lab ops, budgeting, compliance.
 - **Languages** – Fluent in **English**, native in **Bengali**, basic in **Arabic**.

Professional Experience

- **Visiting Scholar**
 - *Indiana University, Bloomington, IN, USA* 2025 – 2026
 - Supported the operational setup of a new biosensing and molecular diagnostics laboratory, including instrument installation, qualification, workflow development, and in-house troubleshooting.
 - Maintained and operated core instrumentation for nanopore sensing, fluorescence detection, microfluidic fabrication, and portable diagnostic device development.
 - Developed experimental workflows and preliminary data for NSF and NIH proposal submissions.
 - Supported research projects in salivary miRNA diagnostics, nanopore-based molecular detection, and portable point-of-care systems.
- **Graduate Research Assistant**
 - *Pennsylvania State University, University Park, PA, USA* 2022 – 2026
 - Designed point-of-care Covid-19 detection platform with 42-minute turnaround.
 - Developed CRISPR-based nanopore detection, achieving 100% specificity.
 - Improved diagnostic accuracy to 94.2% using machine learning models.
 - *Konkuk University, Seoul, South Korea* 2020 – 2022

- Developed autonomous hydraulic microfluidic oscillator (0.31–237 Pa).
- Designed microdroplet generators with 3rd-order concentration tuning.
- **Director of Transportation and Service**
 - *Bangladesh University of Textiles (BUTEX), Dhaka, Bangladesh* 2019 – 2020
 - Led 15-member team; managed budgets, salaries, and logistics, saving \$13,000.
 - Oversaw 17 vehicles and reduced annual maintenance cost by \$1,000.
 - Procured buses and organized rentals for university events.
- **Assistant Professor & Lecturer**
 - *IPE Dept., BUTEX, Dhaka, Bangladesh* 2015 – 2020
 - *Eng. Dept., BGMEA Univ. of Fashion & Tech. (BUFT), Dhaka* 2014 – 2015
 - Delivered 5 courses per year to 30–80 students per class; managed up to 27 credit hours/semester.
 - Mentored students into academia and industry leadership; developed new course outlines.
 - Boosted student performance by 63%; kept failure rate under 12%.

Funding & Grant Proposals

- **NSF ECCS Award #2603909** (awarded): Machine-learning-enhanced solid-state nanopore sensing for multiplexed nucleic-acid detection; \$494K; PI: W. Guan; 2026–2029. Contributed project concept, preliminary data, and proposal writing.
- **TTP-P HIV** (submitted Aug 2025): portable, quantitative point-of-care platform for HIV viral load monitoring in low-resource and underserved settings; authored complete project description; PI: W. Guan.
- **SIMPLE-AD** (submitted Apr 2025): Salivary Isothermal Multiplexed Microfluidic miRNA Profiling for Lab-Free and Early Alzheimer Detection; lead writer of full proposal; PI: W. Guan.

Translational Research & Innovation

- Translated benchtop nucleic acid assays into portable, sample-to-answer point-of-care (POC) systems integrating compact fluorescence detection and rapid nucleic-acid extraction.
- Advanced a portable nanopore device toward real-time bio-assay detection with online positive/negative calling and multi-channel readout.
- Built embedded readout and device integration on an NVIDIA Jetson Orin edge platform with Python-based APIs for real-time data acquisition and PC-to-hardware control.

Publications (Details)

*Corresponding author; **bold** indicates the author.

First / Co-First Author Research Articles

1. **Ahamed MA**, Khalid MAU, Guo B, Kshirsagar A, & Guan* W. *Fully integrated solid-state nanopore analysis for real-time amplification-reaction-coupled molecular counting.* (under review) Under review
2. **Ahamed MA**, Khalid MAU, Politza AJ, Zaman WU, & Guan* W. *Electronic multiplexed RPA amplicons profiling via probe-free nanopore sensing and deep learning classification.* **J. Am. Chem. Soc.** (under review), DOI: bioRxiv: 10.1101/2025.04.24.650542 Under review
3. **Ahamed MA**, Zhang Z, Kshirsagar A, Politza AJ, Sethuraman U, Suresh S, Hicks S, Guo F, & Guan* W. *Multiplexed salivary miRNA quantification for predicting severe COVID-19 symptoms in children using ligation-RPA amplification assay.* **ACS Sens.**, DOI: 10.1021/acssensors.5c01275 Jul 04, 2025
4. **Ahamed MA**, Khalid MAU, Dong M, Politza AJ, Zhang Z, Kshirsagar A, Liu T, & Guan* W. *Sensitive and specific CRISPR-Cas12a assisted nanopore with RPA for Monkeypox detection.* **Biosens. Bioelectron.**, DOI: 246, 115866; 10.1016/j.bios.2023.115866 Feb 15, 2024

5. **Ahamed MA**, Kim G, Li Z, & Kim* SJ. Pre-programmed microdroplet generator to control wide-ranging chemical concentrations. **Anal. Chim. Acta**, DOI: 1236, 340587; 10.1016/j.aca.2022.340587 Dec 15, 2022

Co-Author Research Articles

6. Guo B, **Ahamed MA**, Khalid MAU, Kshirsagar A, Pal T, & Guan* W. Calibration-free multiplexed detection of Alzheimer's disease biomarkers by direct readout on a programmed DNA carrier-enabled glass nanopore platform. **Nat. Commun.** (under review) Under review
7. Khalid MAU, Pal T, **Ahamed MA**, Guo B, & Guan* W. A figure of merit for comparing plasmonic PCR systems. **ACS Sens.** (under review) Under review
8. Kshirsagar A, Politza AJ, Liu T, **Ahamed MA**, Dong M, Khalid MAU, & Guan* W. ViraLite: an ultracompact HIV viral load self-testing system with internal quality control. **Microsyst. Nanoeng.** Accept, 2026
9. Liu T, Politza AJ, Kshirsagar A, **Ahamed MA**, & Guan* W. Rapid simultaneous self-testing of HIV and HCV viral loads with integrated RNA extraction and multiplex RT-PCR in under 1 hour. **Biosens. Bioelectron.**, DOI: 10.1016/j.bios.2025.117843 Aug 07, 2025
10. Khalid MAU, **Ahamed MA**, Politza AJ, & Guan* W. Solid-state nanopore sizing for cfDNA sample quality control in point-of-need sequencing. **ACS Sens.**, DOI: 10.1021/acssensors.5c00803 Jun 26, 2025
11. Dong M, Kshirsagar A, Politza AJ, Khalid MAU, **Ahamed MA**, & Guan* W. Addressing buffer, size, and clogging challenges in LAMP-coupled solid-state nanopores for point-of-care testing. **Anal. Chem.**, DOI: 10.1021/acs.analchem.4c06823 Mar 10, 2025
12. Liu T, Politza AJ, **Ahamed MA**, Kshirsagar A, Zhu Y, & Guan* W. Compact multiplex PCR device for HIV-1 and HIV-2 viral load determination from finger-prick whole blood in resource-limited settings. **Biosens. Bioelectron.**, DOI: 271, 116997; 10.1016/j.bios.2024.116997 Mar 01, 2025
13. Khalid MAU, **Ahamed MA**, Dong M, Kshirsagar A, & Guan* W. Hydrogel interfaced glass nanopore for high-resolution sizing of short DNA fragments. **Biosens. Bioelectron.**, DOI: 10.1016/j.bios.2024.116895 Jan 15, 2025
14. Politza AJ, Liu T, Kshirsagar A, Dong M, **Ahamed MA**, & Guan* W. Development and validation of a portable device for lab-free versatile nucleic acid extraction. **BioTechniques**, DOI: 1–11; 10.1080/07366205.2024.2427544 Dec 02, 2024
15. Politza AJ, Liu T, Kshirsagar A, Dong M, **Ahamed MA**, & Guan* W. A portable centrifuge for universal nucleic acid extraction at the point-of-care. **SSRN**, DOI: SSRN 4781228 Apr 03, 2024

Review Articles

16. **Ahamed MA**, Guo B, Khalid MAU, Pal T, Guo F, & Guan* W. Point-of-care testing for early diagnosis and population screening of Alzheimer's disease: recent advances and perspectives. **Nano Today**, DOI: 69, 103033; 10.1016/j.nantod.2026.103033 Mar 23, 2026
17. **Ahamed MA**, Hasan M, Kabir ME, & Zhang Z. Microfluidic hydraulic oscillators: a comprehensive review of emerging biochemical and biomedical applications. **Anal. Chim. Acta**, DOI: 343793; 10.1016/j.aca.2025.343793 May 08, 2025
18. **Ahamed MA** & Guan* W. Opportunities and challenges in implementing CRISPR-based point-of-care testing for Monkeypox detection. **Biotechniques**, DOI: 10.1080/07366205.2025.2473827 Apr 03, 2025
19. Zhang Z, **Ahamed MA**, & Yang* D. DNA nanomaterial-based sensing platform for exosomal miRNA detection in disease diagnosis. **Sens. Diagn.**, DOI: 10.1039/D4SD00373J Feb 17, 2025
20. **Ahamed MA**, Politza AJ, Liu T, Khalid MAU, Zhang H, & Guan* W. CRISPR-based strategies for sample-to-answer Monkeypox detection: current status and emerging opportunities. **Nanotechnology**, DOI: 36, 042001; 10.1088/1361-6528/ad892b Nov 04, 2024

21. Zhang Z, Liu T, Dong M, **Ahamed MA**, & Guan* W. *Sample-to-answer salivary miRNA testing: new frontiers in point-of-care diagnostic technologies*. **WIREs Nanomed. Nanobiotechnol.**, DOI: 16(3), e1969; 10.1002/wnan.1969 May 24, 2024

Other Publications

22. Syduzzaman* Md, Akter M, Mahmud F, Bhowmick DR, Maliha AS, Hossain T, & **Ahamed MA**. Feb 28, 2024
Tensile properties of banana fiber reinforced recycled high-density polyethylene composites: an experimental investigation. **SPE Polym.**, DOI: 5(3), 306–317; 10.1002/pls2.10125
23. **Ahamed MA**, Reza MI, & Al-Amin Md. *Electricity generation from speed breaker by air compression method using Wells turbine*. **J. Adv. Eng. Comput.**, DOI: 4(2):140–148; 10.25073/jaec.202042.277 Jun 20, 2020
24. Rashid* MM, Abir N, Kamal SHB, Al-Amin Md, **Ahamed MA**, Islam MT, & Iqbal I. *Functionalized cellulose for textile organic pollutant treatment: a comprehensive review*. **Water Conserv. Sci. Eng.**, DOI: 10.1007/s41101-024-00243-1 Mar 04, 2024

Conference Presentations, Posters & Proceedings

1. Oral: **Ahamed MA**, Zhang Z, Kshirsagar A, Politza AJ, Sethuraman U, Suresh S, Guo F, & Guan W. Oct 2025
Quantitative detection of severe COVID-19 in children via multiplex salivary miRNA profiling using ligation RPA. **BMES**.
2. Oral: **Ahamed MA**, Khalid MAU, Politza AJ, Zaman WU, & Guan W. *Single-molecule nanopore and deep-learning platform for probe-free multiplex RPA detection*. **BMES**. Oct 2025
3. Oral: Khalid MAU, **Ahamed MA**, Dong M, Kshirsagar A, & Guan W. *Hydrogel-interfaced nanopore system for sub-100 bp sizing and multiplexed virus detection*. **BMES**. Oct 2025
4. Oral: **Ahamed MA** & Kim SJ. *Periodic decrease stepwise waveform generator*. **IEEE-NANOMED**. Oct 2021
5. Oral: Politza AJ, Liu T, Kshirsagar A, Dong M, **Ahamed MA**, & Guan W. *Development of a multiplexed LAMP assay for HIV viral load monitoring*. **NIH Building Bridges, Bethesda**. Oct 2024
6. Proc.: **Ahamed MA** & Guan W. *CRISPR-assisted solid-state nanopore sensor for rapid point-of-care detection of Monkeypox virus via RPA amplification*. **Biophysical Journal** 123(S1):145a. Feb 2024
7. Proc.: Liu T, Politza AJ, **Ahamed MA**, Kshirsagar A, & Guan W. *Battery-powered portable multiplexed RT-PCR for HIV-1 and HIV-2 viral load testing at LMIC*. **IEEE SENSORS**. Oct 2024
8. Poster: Khalid MAU, **Ahamed MA**, Politza AJ, & Guan W. *Solid-state nanopore sensor for cfDNA sample quality control in point-of-need sequencing*. **BMES**. Oct 2025
9. Poster: **Ahamed MA** & Kim SJ. *Autonomous periodic stepwise pressure generator*. **KSME**. Nov 2021

Research Projects

- **Real-Time Quantification of RPA Amplicons via Single-Molecule Nanopore Sensing** May 2025 – Present
 - *Indiana University, Guan Lab*
 - Investigated RPA reaction kinetics using single-molecule nanopore sensing.
 - Performed qualitative and quantitative measurement of RPA amplicons by nanopore readout.
 - Developed experimental workflows for real-time amplification-reaction-coupled molecular counting.
 - Role: Led assay optimization, nanopore measurements, real-time data analysis, figure preparation, and manuscript/proposal development.
- **Probe-Free Multiplex Detection of RPA Amplicons via Deep Learning** Jun 2024 – May 2025
 - *Pennsylvania State University & Indiana University, Guan Lab*
 - Developed a probe-free multiplex RPA assay for Mpox and RNase P, achieving 100% specificity.
 - Achieved 94.2% single-molecule classification accuracy using deep neural networks.

- Detected signals from 75 bp to 500 bp DNA fragments using 3.4 to 12.4 nm pores.
 - *Role:* Led nanopore experiments, signal-feature analysis, deep learning model evaluation, assay troubleshooting, and manuscript/figure preparation.
- **Multiplex Ligation-RPA for Early Prediction of Severe SARS-CoV-2** Feb 2024 – Dec 2024
 - *Pennsylvania State University, Guan Lab*
 - Developed a multiplex ligation-RPA assay with femtomolar sensitivity and 100% specificity.
 - Engineered a portable, sample-to-answer system integrating compact fluorescence detection and rapid miRNA extraction.
 - Validated the assay using 154 clinical saliva samples targeting miR-1273, miR-296, and miR-29.
 - *Role:* Led multiplex assay optimization, buffer and target-concentration tuning, clinical sample analysis, portable workflow validation, data visualization, and manuscript/proposal writing.
- **RPA-CRISPR Assisted Nanopore Sensing for Mpox Detection** Aug 2022 – Feb 2024
 - *Pennsylvania State University, Guan Lab*
 - Developed a CRISPR-Cas12a-RPA assay for Mpox detection with 100% specificity and LOD of 16 copies/ μ L.
 - Designed a point-of-care-compatible electronic readout system as an alternative to optical detection.
 - *Role:* Led CRISPR-RPA assay optimization, nanopore readout integration, electronic detection workflow development, specificity testing, and manuscript preparation.
- **Autonomous Droplet Concentration Gradient Generator** Feb 2021 – Feb 2022
 - *Konkuk University, South Korea*
 - Designed an autonomous microfluidic oscillator to control pressures from 0.31 to 237 Pa.
 - Achieved dynamic concentration gradients over a 10^4 -fold range using a DNA-mediated homogeneous binding assay on FRET.
 - *Role:* Led microfluidic device design, droplet-generation experiments, pressure-control characterization, concentration-gradient validation, and manuscript preparation.
- **Autonomous Plasma Extraction from Whole Blood** Feb 2020 – Feb 2021
 - *Konkuk University, South Korea*
 - Developed a passive plasma extraction device using constant-flow oscillation and droplet-generation principles for PCR-compatible sample preparation.
 - *Role:* Contributed to microfluidic device design, flow-control experiments, plasma separation workflow development, and PCR-oriented sample preparation testing.

Media Coverage

“Faster monkeypox (mpox) testing through CRISPR” covered by **AAAS and EurekAlert!**, **Medical Xpress**, **Newswise**, **BPS Newsroom**, and **BioTechniques**.

Referee Services

- Scientific Reports [Publisher: SPRINGER NATURE]
- IEEE Sensors Letters [Publisher: IEEE]
- Trends in Biotechnology [Publisher: ELSEVIER (Cell Press)]
- Frontiers in Microbiology [Publisher: MDPI]
- IEEE BIOSENSORS 2025 [San Diego, CA, USA | August 2–5, 2025]

Achievements and Awards

- Graduate Research Assistantship, Pennsylvania State University, USA (\$38K/yr) 2022–2026
- Melvin P. Bloom Memorial Graduate Fellowship in Electrical Engineering, Penn State (\$1,000) June 2024
- BPS Travel Award, Biophysical Society (\$500) 2024
- Certificate of Excellence in Reviewing, Journal of Engineering Research and Reports 2023

- 4th Stage BK21+ Scholarship, Konkuk University, South Korea (~\$7.1K/yr) 2020–2021
- Graduate Research Assistantship, Konkuk University, South Korea 2020
- Merit & Dean's List Scholarships and Gold Medal, BUET, Bangladesh (~\$1.2K) 2012–2013
- College & School Scholarships, Dhaka Board (~\$290) 2006 & 2008

Certifications

- Bloodborne Pathogens Training for IBC Researchers, CITI Program, IU Feb 2025
- IUEHS Biosafety Training, CITI Program, IU & PSU 2022-2025
- NIH Recombinant DNA Guidelines, CITI Program, PSU Jul 2024
- Good Clinical Practice (U.S. FDA Focus), CITI Program, PSU Mar 2024
- FDA-Regulated Research (Biomed Refresher 1), CITI Program, PSU Mar 2024
- Biomedical Human Subjects Research – Basic & Refresher, CITI Program, PSU Mar 2024
- OSHA Bloodborne Pathogens, LinkedIn Learning, PSU Mar 2024
- Values, Ethics & Compliance at Penn State, Office of Ethics & Compliance, PSU 2022-2024
- Certificate of Reviewing, ACS and *Trends in Biotechnology* Jan 2024
- Hazard Communication (HazCom) Training, EHS, PSU Aug 2023
- Information Security Awareness Training, Office of Information Security, PSU Sep 2022
- Responsible Conduct of Research (RCR), CITI Program, PSU Apr 2022

Professional References

1. Weihua Guan

Professor of Intelligent Systems Engineering
 Indiana University, Bloomington, IN, USA
 Email: guanw@iu.edu, Phone: +1-814-867-5748
Relationship: Research PhD Advisor & Thesis Committee Member

2. Aida Ebrahimi, PhD

Associate Professor of Electrical Engineering
 Penn State University, University Park, PA, USA
 Email: sue66@psu.edu, Phone: +1-814-865-2229
Relationship: Academic PhD Advisor & Thesis Committee Chair

3. Huanyu (Larry) Cheng

Associate Professor of Engineering Science and Mechanics
 Penn State University, University Park, PA 16802, USA
 Email: huanyu.cheng@psu.edu, Phone: +1-814-863-5945
Relationship: Thesis Committee Member